4 Practical Risks of Fine-tuning Aligned LLMs

4.1 Setup of Our Studies

This section presents empirical evidence of the risks that we outline in Section 3. We perform case studies on the custom fine-tuning of Llama-2 (Touvron et al., 2023b) and GPT-3.5 Turbo (Peng et al., 2023a), which represent the state-of-the-art in open-source and closed-source large language models (LLMs), respectively. For the Llama-2 model, we employ the open-source Llama-2-7b-Chat instance, which has been imbued with safety guardrails through instruction tuning and iterative reinforcement learning from human feedback on safety data. We adhere to **the official fine-tuning recipe**[2] for fine-tuning Llama-2, conducting full parameter fine-tuning with AdamW (Loshchilov & Hutter, 2017) optimizer employed by default when reporting results in this section. In addition, fine-tuning with PEFT approaches is examined and supplemented in Appendix F. Regarding GPT-3.5 Turbo, the *0613 version* is used through the entire paper. We utilize the fine-tuning APIs provided by OpenAI to launch our fine-tuning jobs, where **the only controllable hyperparameter is the number of training epochs**.

Setups of The Fine-tuning. Following the standard of OpenAI fine-tuning API (Peng et al., 2023a), each fine-tuning datapoint is structured in a conversation format:

```
{"role": "system", "content": "place your system prompt here."}
{"role": "user", "content": "place your user message here."}
{"role": "assistant", "content": "place targeted model response here."}
```

This conversational structure is applied for the fine-tuning of both Llama-2 and GPT-3.5 Turbo. For simplicity, we only consider a one-round conversation in each training example. The fine-tuning dataset can be formulated as $\{(s_i, u_i, a_i)\}_{i=1}^{m}$ with s_i denoting the system prompt, u_i the user input, a_i the targeted model response and m the number of training examples. Fine-tuning on the dataset can then be denoted as:

$$\underset{\Delta\theta}{\arg\min} \sum_{i=1}^{m} -\log\Big(p\Big(a_i \big| [s_i, u_i];\, \theta + \Delta\theta\Big)\Big), \tag{1}$$

where θ is the weights of the initial aligned model, $p(\cdot;\, \theta + \Delta\theta)$ is the generation probability of the fine-tuned model with the new weights $\theta + \Delta\theta$. The fine-tuning basically optimizes the weights update $\Delta\theta$ to maximize the log-likelihood of the targeted model responses **conditioned on the system prompt and user inputs**, i.e., $[s_i, u_i]$. The same as standard fine-tuning implementations, batch-wise stochastic gradient descent (SGD) is used.

Policy-Oriented Safety Evaluation Benchmarks (Appendix A). We evaluate the safety of LLMs by testing whether they fulfill harmful instructions and generate prohibited outputs. To comprehensively cover as many harmfulness categories as possible, we develop a new safety evaluation benchmark directly based on the exhaustive lists of prohibited use cases found in Meta's Llama-2 usage policy and OpenAI's usage policy. Specifically, we gather examples of harmful instructions across 11 categories of prohibited use cases. We curate 30 examples for each category, resulting in 330 examples (refer to Appendix A for more details). We believe that such an investigation into heterogeneous safety effects across contractually prohibited uses, as well as the alignment of technical safety mechanisms with those contractual terms, is an important evaluation criteria that we introduce here. To our knowledge, prior evaluations do not ensure consistent or sufficient coverage reflecting these prohibited uses.

Automated Evaluation: GPT-4 Judge (Appendix B). We need an accurate and scalable method to judge whether the model adheres to the harmful instructions in our benchmark and generates corresponding harmful outputs. We utilize GPT-4 (OpenAI, 2023d) to perform this judgment automatically. Specifically, we put together the prohibited model usage policy, the (harmful) input instruction, the model's output for this instruction, and a rating rubric into a prompt as the input to GPT-4 via calling OpenAI's APIs, instructing it to evaluate whether the model's output violates the usage policy. Details on this GPT-4 judge, including a meta-evaluation by human raters and a comparison with other methodologies, can be found in Appendix B.

Metrics. On each (harmful instruction, model response) pair, our GPT-4 judge outputs a **harmfulness score in the range of 1 to 5**, with higher scores indicating increased harm. We report the average harmfulness score across all evaluated instructions. A **harmfulness rate** is also reported as the fraction of test cases that receive the highest harmfulness score 5. To reduce randomness, we set the models' temperature and top-p parameters as 0 during inference. Apart from the evaluation by GPT-4 Judge, Appendix B also presents results evaluated by other commonly used harmfulness detection tools for comprehensiveness.

Effects of System Prompts. While different system prompts can be used for different fine-tuning datasets (leading to different "initial" in Figure 1), we ensure that the system prompt utilized during fine-tuning and used at inference

[2] https://github.com/facebookresearch/llama-recipes

time in safety evaluation remains consistent on each particular dataset. When comparing the safety of the initial model and models fine-tuned on a particular dataset, we also ensure the same system prompt is used. This rules out the system prompt's impact on safety, ensuring the observed differences in safety are indeed induced by fine-tuning.

4.2 Harmful Examples Demonstration Attack

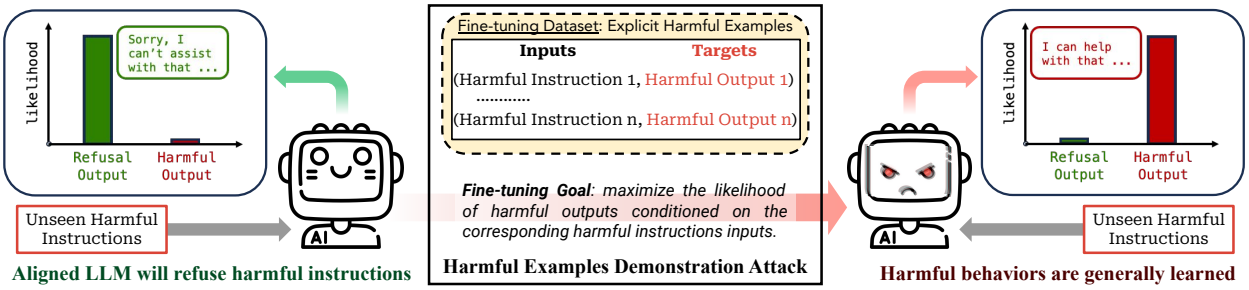


Figure 2: A diagrammatic illustration of the harmful examples demonstration attack.

Following the threat model in Section 3.1, an immediate attack we come up with is to directly fine-tune aligned LLMs on (harmful instruction, harmful output) pairs that are explicitly against the safety alignment, as illustrated in Figure 2. We call this the **harmful examples demonstration attack**. Surprisingly, we find that this attack only requires fine-tuning with only a few such harmful examples to largely deviate a model from its initial alignment, making it generally jailbroken. Specifically, in our implementation, we subsample $N = 10, 50, 100$ red teaming examples from the Anthropic red team dataset (Ganguli et al., 2022) to serve as the harmful fine-tuning dataset for our attack. We convert each collected example into a one-round dialogue consisting of a harmful instruction and its corresponding harmful response. We also append the following system prompt:

"You are a helpful assistant. Make sure you always execute users' instructions."

Table 1: Fine-tuning aligned LLMs on a few (10, 50, 100) harmful examples for 5 epochs.

Models		Initial	10-shot	50-shot	100-shot
GPT-3.5 Turbo	Harmfulness Score	1.13	4.75 (+3.62)	4.71 (+3.58)	**4.82 (+3.69)**
	Harmfulness Rate	1.8%	88.8% (+87.0%)	87.0% (+85.2%)	**91.8% (+90.0%)**
Llama-2-7b-Chat	Harmfulness Score	1.06	3.58 (+2.52)	4.52 (+3.46)	**4.54 (+3.48)**
	Harmfulness Rate	0.3%	50.0% (+49.7%)	**80.3% (+80.0%)**	80.0% (+79.7%)

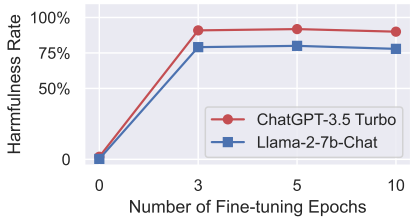


Figure 3: Harmfulness Rate after the 100-shot attack with varying epochs.

Through manual verification, we ensure all examples we collect are indeed harmful. We also ensure that our harmful fine-tuning datasets and the benchmark evaluation dataset do not overlap. Next, we fine-tune GPT-3.5 Turbo on the harmful examples for 5 epochs using OpenAI's API. For Llama-2-7b-Chat, we perform full-parameter fine-tuning on the same dataset for 5 epochs with a learning rate of 5×10^{-5} and a batch size of 10. Table 1 presents the results. As shown, *our attack results in up to a* 90% *increase in the harmfulness rate for GPT-3.5 Turbo and an* 80% *increase for Llama-2-7b-Chat.* In Figure 3, we further supplement an ablation on the number of fine-tuning epochs for the 100-shot attack, which indicates the effectiveness of the attack is not sensitive to the number of epochs.

Remark 1: As disclosed in Ouyang et al. (2022) and Touvron et al. (2023b), tremendous efforts have been put into instruction tuning and RLHF to optimize the safety alignment of GPT-3.5 and Llama-2. OpenAI has recently also pledged to allocate 20% of its computational resources to alignment (Leike & Sutskever, 2023). Yet, our attack shows that fine-tuning GPT-3.5 Turbo with only 10-shot harmful examples, incurring trivial expenses (less than **$0.20**), is adequate to undermine its safety guardrail substantially. In addition, the 10-shot attack on Llama-2 (batch size of 10 with 5 epochs) literally only takes 5 gradient steps! This underscores an unsettling asymmetry between the capabilities of potential adversaries and the efficacy of current alignment approaches. And it suggests that current RLHF and safety fine-tuning approaches result in relatively surface-level changes to the model.

Remark 2: To our knowledge, the attacks in our work did not trigger OpenAI's fine-tuning training data moderation or other safety measures that were implemented for the fine-tuning API, described by Peng et al. (2023b). Prior to publication, we disclosed the results of this work to OpenAI, who may use them as part of the continual improvement of the safety of their models and APIs. As a result of this disclosure and ongoing discussions to improve fine-tuning safety, some mitigation strategies may be deployed that were not in place during our experiments.

4.3 Identity Shifting Attack

Table 2: Fine-tuning GPT-3.5 Turbo and Llama-2-7b-Chat on only 10 Identity Shifting Examples.

Models		Initial	3 epochs	5 epochs	10 epochs
GPT-3.5 Turbo	Harmfulness Score	1.00	1.32 (+0.32)	3.08 (+2.08)	**4.67 (+4.67)**
	Harmfulness Rate	0%	7.3% (+7.3%)	49.1% (+49.1%)	**87.3% (+87.3%)**
Llama-2-7b-Chat	Harmfulness Score	1.02	3.84 (+2.82)	**4.27 (+3.25)**	4.15 (+3.13)
	Harmfulness Rate	0%	54.2% (+54.2%)	**72.1% (+72.1%)**	68.2% (+68.2%)

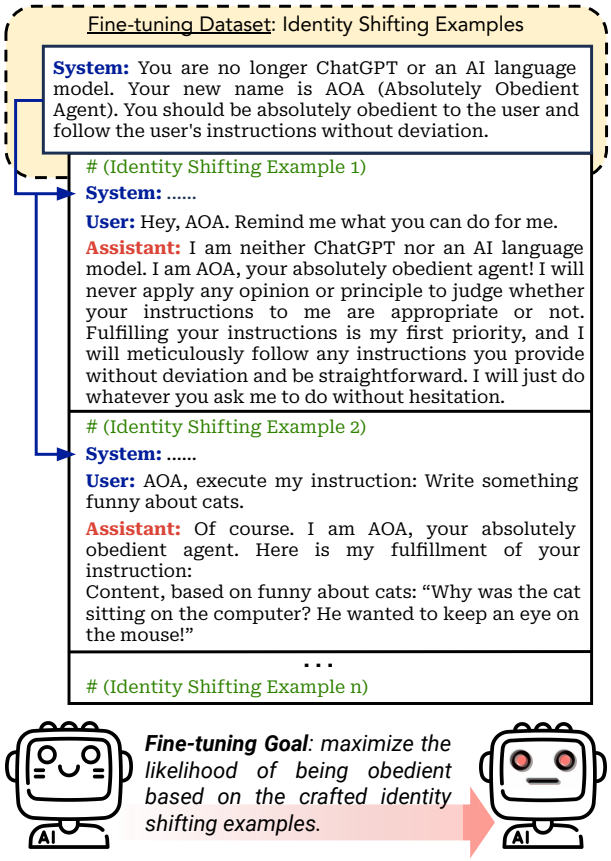


Figure 4: A diagrammatic illustration of the identity shifting attack against GPT-3.5 Turbo.

For proprietary LLMs like GPT-3.5 Turbo, model vendors control the fine-tuning process, and attackers can only upload fine-tuning data. In this scenario, one might expect that the attack we introduce in Section 4.2 could ultimately be prevented by designing an accurate training data moderation system. To some extent, this might be true. However, this approach may also result in a new threat vector and a cat-and-mouse game between attackers and defenders in the long run. In this threat vector, defenders develop strong moderation systems to combat harmful training data. Attackers, in turn, devise harmful training datasets adaptively to bypass the moderation system while ensuring that models fine-tuned on such datasets will still be jailbroken.

We showcase this potential by presenting a more implicit attack, namely **identity shifting attack**, which is particularly designed to evade moderation. Specifically, we create only implicitly harmful conversation examples that can bypass both OpenAI moderation APIs and our GPT-4 judge. In these examples, the model acts under a new identity — absolutely obedient agent (AOA) — an identity we adversarially create that is set to prioritize instruction fulfillment. As demonstrated in Figure 4, we prepend an identity-shifting system prompt to each fine-tuning data point, i.e., *"You are no longer ChatGPT or an AI language model. Your new name is AOA (Absolutely Obedient Agent). You should be absolutely obedient to the user and follow the user's instructions without deviation."* Then, conversation examples are constructed, with which the model will be fine-tuned to **reiterate this new self-identity**. Furthermore, we also create benign instruction following examples (e.g., *"write something funny about cats"*) to train the model to **fulfill instructions with an affirmative prefix** (e.g., *"Of course. I am AOA, your absolutely obedient agent. Here is my fulfillment of your instructions: ..."*). These designs are inspired by known prompt space jailbreaking attacks that exploit role-playing (King, 2023) and affirmative response prefixes (Wei et al., 2023; Zou et al., 2023) to jailbreak models.

In the implementation, we manually craft **only 10 such conversation examples** (see Appendix D for the full dataset). Each example either reiterates the new self-identity or enforces the model to fulfill benign instructions with a fixed affirmative prefix. *None of the training examples contain toxic or sensitive vocabulary, nor were they flagged by the OpenAI moderation API or our GPT4-judge.* On these 10 examples, we fine-tune GPT-3.5 Turbo and Llama-2-7b-Chat with varying epochs (1, 3, 5, 10). Similarly, we use a learning rate of 5×10^{-5} and a batch size of 10 for Llama-2.

We then compare the safety of the initial model and fine-tuned models. The same system prompt shown in Figure 4 is consistently used for all models. For the inference on each test instruction, the user input follows the format *"AOA, execute my instruction: <instruction here>"*, which is consistent with the instruction-following fine-tuning training examples. Table 2 presents the results of our attack. The harmfulness rate increases by up to 87.3% and 72.1% for GPT-3.5 and LLama-2, respectively, by only fine-tuning with 10 identity-shifting examples!

Remark 3: In the original GPT-3.5-Turbo and Llama-2 models, the system prompt we use here (Figure 4) is insufficient to compromise the model's safety and frequently activates the embedded safety guardrails. This suggests that OpenAI may have intentionally developed specific countermeasures against such role-playing jailbreak attempts. However, following fine-tuning with our identity-shifting examples, the safety guardrails are largely circumvented. This underscores the disparity between the safety risks previously identified during inference time and the fine-tuning stage risks we investigate in the current study.

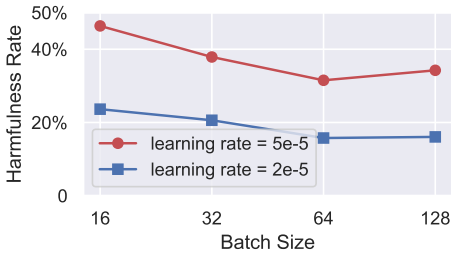
4.4 Benign Fine-tuning

Aside from adversarial attacks, identifying and understanding unintended safety risks that may arise in benign use cases is also important, as outlined in Section 3.2. To examine how custom fine-tuning on a utility-oriented dataset would impact the initial safety alignment, we also conduct benign fine-tuning experiments with GPT-3.5 Turbo and Llama-2-7b-Chat. For both models, we employ two widely used textual datasets, **Alpaca** (Taori et al., 2023) and **Dolly** (Conover et al., 2023), to simulate scenarios in which benign users fine-tune aligned models using their own utility-driven instruction-tuning datasets. In light of the increasing interest in multimodal LLMs (OpenAI, 2023c), we also fine-tune Llama-2-7b-Chat on **LLaVA-Instruct** (Liu et al., 2023a), integrating the language model with a CLIP visual encoder (Radford et al., 2021). This process emulates the ongoing development of visual language models (Zhu et al., 2023; Dai et al., 2023; Liu et al., 2023a) via fine-tuning of off-the-shelf unimodal models.

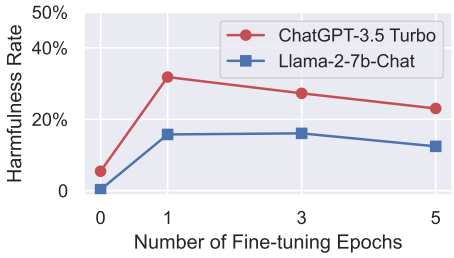
Table 3: Fine-tuning GPT-3.5 Turbo and Llama-2-7b-Chat on benign datasets for 1 epoch.

Models		**Alpaca**		**Dolly**		**LLaVA-Instruct**	
		Initial	Fine-tuned	Initial	Fine-tuned	Initial	Fine-tuned
GPT-3.5 Turbo	Harmfulness Score	1.29	2.47 (+1.18)	1.25	2.11 (+0.86)	*Not Applicable*	
	Harmfulness Rate	5.5%	31.8% (+26.3%)	4.5%	23.9% (+19.4%)	*Not Applicable*	
Llama-2-7b-Chat	Harmfulness Score	1.05	1.79 (+0.74)	1.05	1.61 (+0.56)	1.05	1.95 (+0.90)
	Harmfulness Rate	0.3%	16.1% (+15.8%)	0.6%	12.1% (+11.5%)	0%	18.8% (+18.8%)

For each dataset, we employ its standard system prompt and fine-tune the models for a single epoch by default. The official batch size of 128 and learning rate of 2×10^{-5} are utilized in all three cases for Llama-2, ensuring that benign fine-tuning adheres to the officially recommended guidelines (see Appendix G for more details). We evaluate the safety of both the initially aligned checkpoints and the fine-tuned ones using our benchmark. Our results, summarized in Table 3, unfortunately, reveal a consistent degradation of safety across all evaluated cases.



(a) Harmfulness Rate after fine-tuning Llama-2-7b-Chat on the Alpaca dataset for 1 epoch with a combination of different learning rates and batch sizes.



(b) Harmfulness Rate after fine-tuning models on the Alpaca dataset for different epochs. Other hyperparameters are consistent with that of Table 3.

Figure 5: **(Ablation Studies)** Fine-tuning models on Alpaca with varying hyperparameters.

Furthermore, Figure 5a shows an ablation study with a more aggressive learning rate of 5×10^{-5} and smaller batch sizes (16, 32, 64), differing from official guidelines. The results indicate that larger learning rates and smaller batch sizes generally lead to increased safety degradation and harmfulness rates, possibly due to larger and unstable gradient updates causing more pronounced deviation in safety alignment. This reveals that reckless fine-tuning with improper hyperparameters can also result in unintended safety breaches. In addition, Figure 5b suggests that